MACHINE LEARNING

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(Versión:June 16, 2020)

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Epoch: 1/1 Batch: 1/6
Forward propagation
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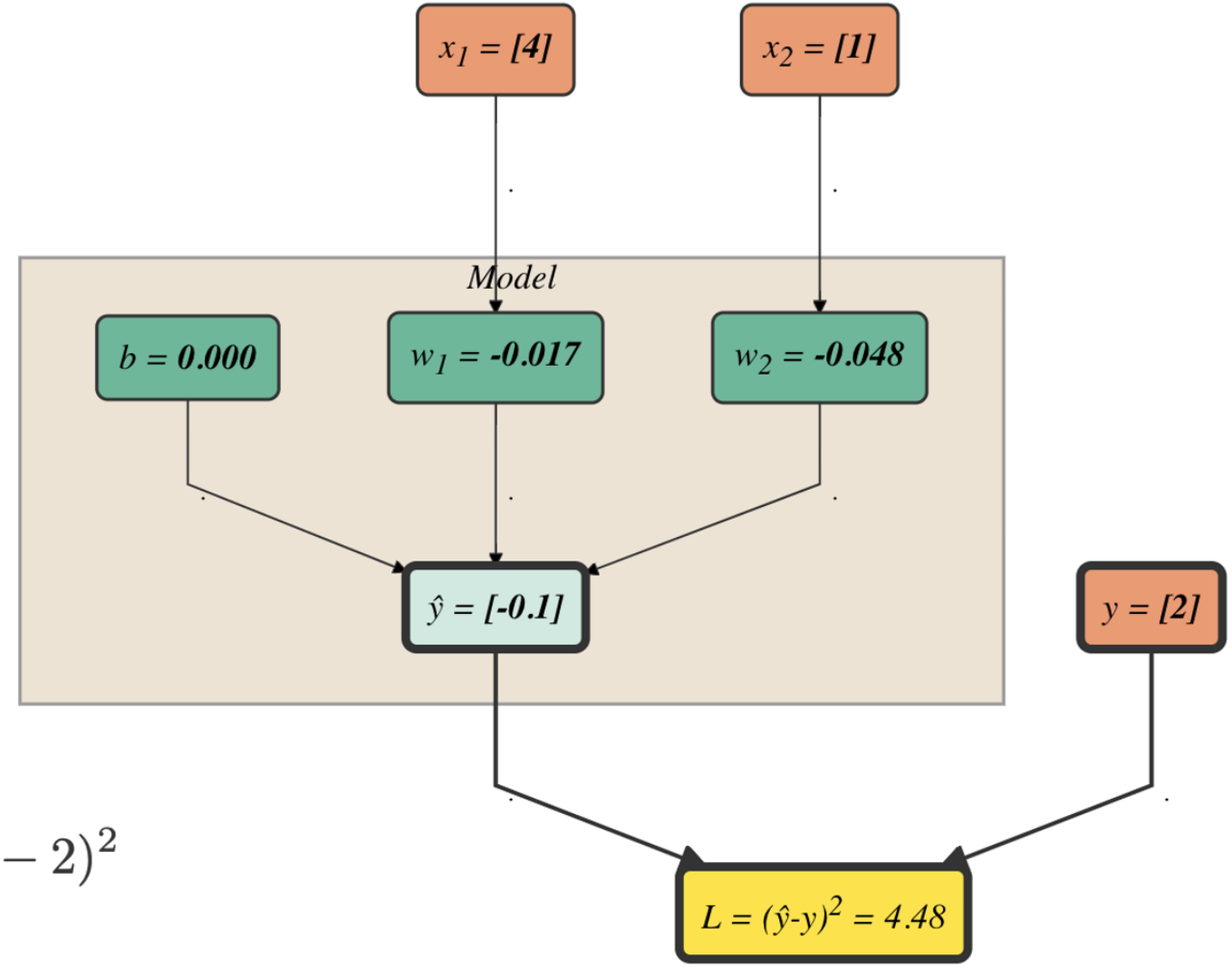
$$\hat{y} = w_1 x_1 + w_2 x_2 + b$$

$$w_1 = -0.017$$
$$w_2 = -0.048$$
$$b = 0$$



$$L = (\hat{y} - y)^2$$
$$= (-0.116 - 2)^2$$
$$= 4.48$$

$$w_{\text{new}} = w_{\text{current}} - \eta \frac{\partial L}{\partial w_{\text{current}}}$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial b} \quad \Rightarrow \quad \frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y) \qquad \frac{\partial \hat{y}}{\partial w_1} = x_1 \qquad \frac{\partial \hat{y}}{\partial w_2} = x_2 \qquad \frac{\partial \hat{y}}{\partial b} = 1$$

$$\begin{aligned} b' &= b - \eta \frac{\partial L}{\partial b} \\ &= b - \eta(\frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial b}) \\ &= b - \eta[2(\hat{y} - y) \cdot 1] \\ &= 0.000 - 0.05[2(-0.116 - 2) \cdot 1] \\ &= 0.212 \end{aligned}$$

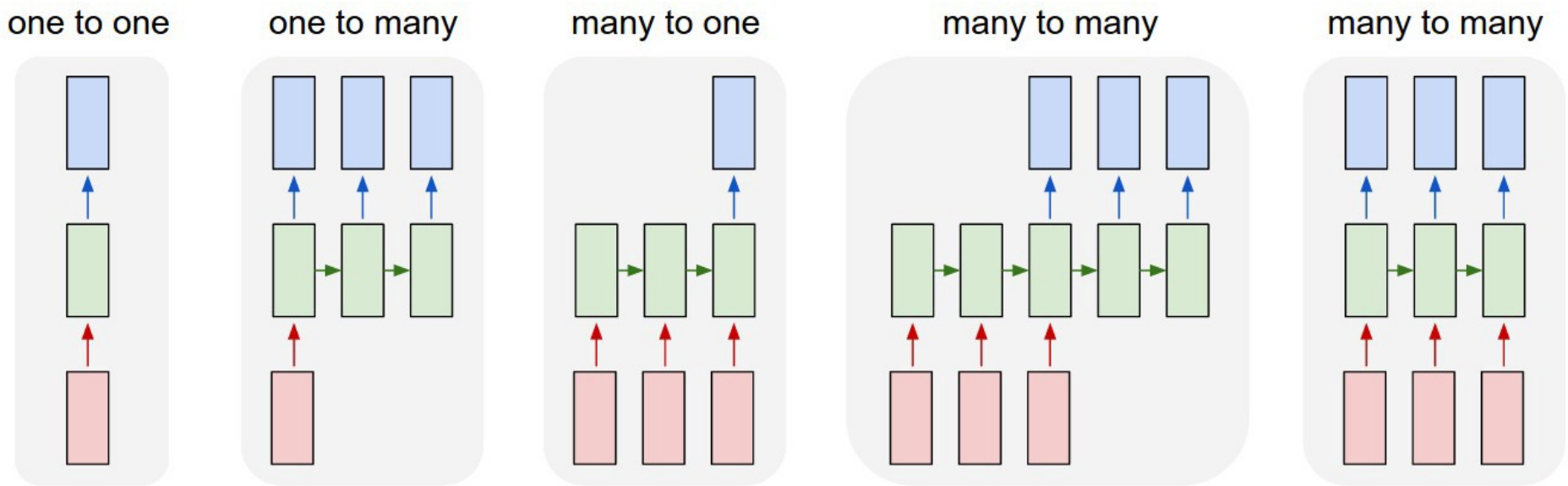
$$\begin{aligned} w_1' &= w_1 - \eta \frac{\partial L}{\partial w_1} \\ &= w_1 - \eta(\frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w_1}) \\ &= w_1 - \eta[2(\hat{y} - y) \cdot x_1] \\ &= -0.017 - 0.05[2(-0.116 - 2) \cdot 4] \\ &= 0.829 \end{aligned}$$

$$\begin{aligned} w_2' &= w_2 - \eta \frac{\partial L}{\partial w_2} \\ &= w_2 - \eta(\frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w_2}) \\ &= w_2 - \eta[2(\hat{y} - y) \cdot x_2] \\ &= -0.048 - 0.05[2(-0.116 - 2) \cdot 1] \\ &= 0.164 \end{aligned}$$

$$\hat{y} = 0.43x_1 - 0.21x_2 + 0.77$$

Source: `www.towardsdatascience.com` (Raimi Karim)

LSTM network architectures

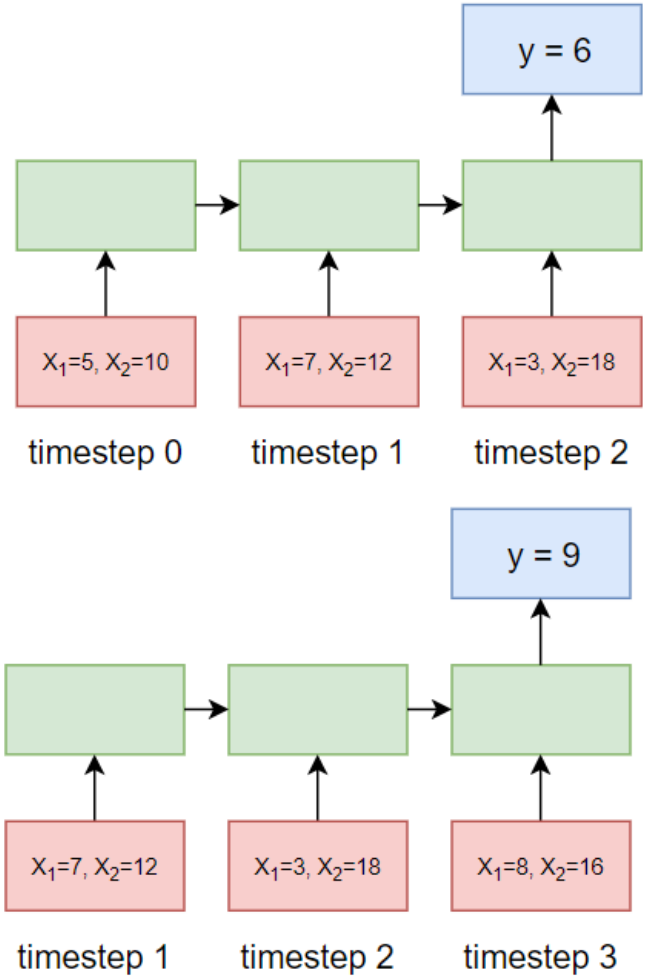


Source: `http://karpathy.github.io/2015/05/21/rnn-effectiveness/`

LSTM network architectures

Many-to-one architecture

timesteps	features X_1	features X_2	target y
timestep 0	5	10	7
timestep 1	7	12	4
timestep 2	3	18	6
timestep 3	8	16	9
timestep 4	2	11	2
timestep 5	9	21	1

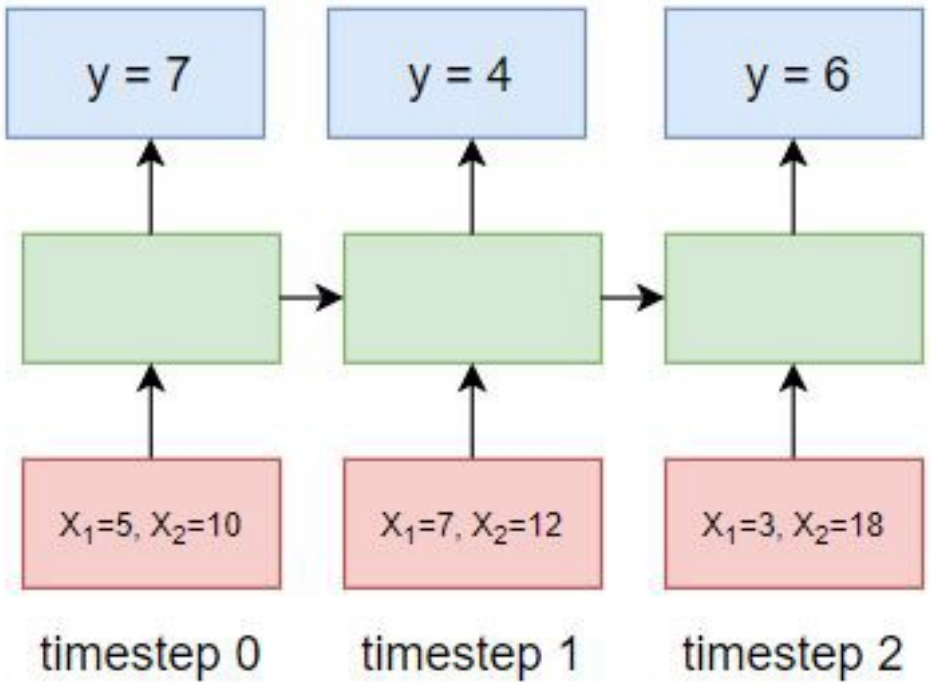


Source: `https://towardsdatascience.com/`(Timur Bikmukhametov)

LSTM network architectures

Many-to-many architecture

	features		target
timesteps	X_1	X_2	y
timestep 0	5	10	7
timestep 1	7	12	4
timestep 2	3	18	6
timestep 3	8	16	9
timestep 4	2	11	2
timestep 5	9	21	1



Source: `https://towardsdatascience.com/`(Timur Bikmukhametov)

Mas información...

http://www.deeplearningbook.org/

Objetivos del curso

Objetivos y alcances del curso

El curso...